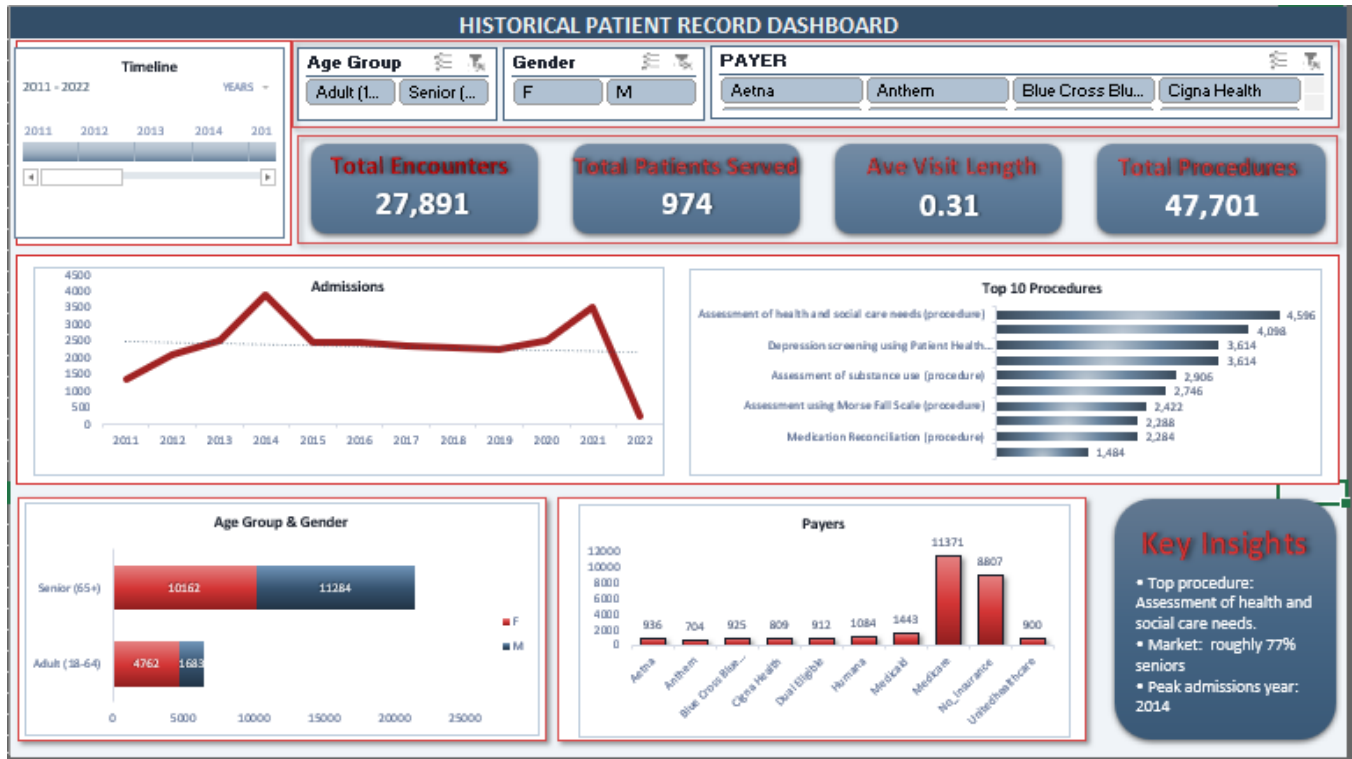


## EXCEL PORTFOLIO CASE STUDY

# Historical Patient Record Dashboard

From fragmented patient activity to an executive-ready view of utilization, patient mix, procedures, payers, and longitudinal admissions.



Dashboard snapshot supplied for this portfolio case study.

| TOTAL ENCOUNTERS | PATIENTS SERVED | AVG. VISIT LENGTH | TOTAL PROCEDURES |
|------------------|-----------------|-------------------|------------------|
| 27,891           | 974             | 0.31              | 47,701           |

## Portfolio positioning

This project demonstrates the ability to turn a historical healthcare dataset into a decision-oriented Excel dashboard. The dashboard combines KPI design, time-series analysis, demographic segmentation, payer analysis, procedure utilization, and an executive insight layer.

## 1. Project Overview

The Historical Patient Record Dashboard is an executive analytics solution designed to answer a practical healthcare operations question: What does the historical patient population look like, how is care being utilized, and where should leadership focus attention?

The dashboard covers a 2011–2022 historical period and brings together patient encounters, age/gender composition, payer mix, admissions by year, and the most frequently recorded procedures. The design intentionally moves from high-level KPIs to diagnostic visuals and ends with a concise Key Insights panel so a decision-maker can understand the story without navigating a large report.

### Problem statement

Healthcare records can contain large volumes of transactional activity that are difficult to interpret when viewed as raw tables. Leadership needs a compact analytical layer that highlights utilization patterns, population characteristics, and operational priorities.

### Analytical objective

- Create a single source of truth for historical patient activity and utilization.
- Quantify encounters, patients served, visit length, and procedure volume.
- Identify changes in admissions over time and surface the peak year.
- Profile the population by age group and gender.
- Compare payer categories and identify concentration or coverage gaps.
- Rank procedures to identify the highest-volume clinical activities.
- Translate descriptive analytics into practical operational recommendations.

### Scope and data caveat

This case study is based on the values shown in the Excel dashboard rather than a re-extraction of the underlying source tables. Therefore, the figures shown here are treated as dashboard-observed values. Recommendations and impact estimates are proposed opportunities, not claims of realized financial or clinical outcomes.

## 2. Business Questions

| Business Lens    | Question                                                                                                   |
|------------------|------------------------------------------------------------------------------------------------------------|
| Utilization      | How many encounters and procedures are being recorded, and what is the average visit length?               |
| Demand trend     | How do admissions change over time, and which year represents the highest observed demand?                 |
| Population mix   | What proportion of the population is senior versus adult, and how does the gender mix differ by age group? |
| Payer mix        | Which payers represent the largest observed volumes, and how significant is the uninsured segment?         |
| Procedure demand | Which procedures account for the largest volumes and therefore deserve capacity or workflow                |

**Decision support**

attention?

Can the dashboard surface the most important findings without requiring users to manually interpret every chart?

### 3. Key Insights

The dashboard supports several portfolio-worthy insights. The strongest findings are summarized below.

#### Senior population is the dominant segment

The age/gender visual shows 21,446 senior records versus 6,445 adult records, meaning seniors represent approximately 76.9% of the displayed population. The dashboard itself summarizes this as roughly 77% seniors.

#### 2014 is the peak admissions year

The admissions trend reaches its highest point in 2014, making that year a useful benchmark for demand, staffing, capacity, and service-mix analysis.

#### Procedure demand is concentrated

The leading procedure is Assessment of Health and Social Care Needs, with approximately 4,596 recorded instances in the dashboard. Other high-volume activities include depression screening, substance-use assessment, Morse Fall Scale assessment, and medication reconciliation.

#### Payer mix is highly uneven

The payer chart shows much larger volumes for Medicare and No Insurance than for the named commercial payers. This suggests that payer concentration and coverage status should be monitored alongside utilization.

#### High activity relative to patient count

With 27,891 encounters across 974 patients, the dashboard implies substantial repeat utilization. A simple encounter-to-patient ratio is about 28.6 encounters per patient, although this should be validated against the exact grain and inclusion rules in the underlying data.

#### The dashboard is designed for fast executive scanning

The combination of four KPI cards, one trend visual, ranked procedures, demographic segmentation, payer distribution, and a Key Insights panel creates a strong narrative hierarchy.

#### Important interpretation note

The 28.6 encounter-to-patient ratio is a derived portfolio metric calculated from the displayed KPIs ( $27,891 \div 974$ ). It is not a claim about unique patients' clinical journeys; the source grain and patient-level inclusion rules should be validated before operational use.

## 4. Business Recommendations

The recommendations below are directed at the hospital facility and its clinical, operational, and financial leadership. They translate the dashboard findings into actions that could improve patient care, capacity planning, safety, and financial sustainability.

### 1. Build a senior-centered care program

Seniors account for approximately 77% of the displayed population. The hospital should strengthen geriatric-focused care pathways, including fall-risk assessment, medication reconciliation, social-needs screening, discharge planning, and care coordination.

### 2. Strengthen fall-prevention and patient-safety protocols

The high volume of Morse Fall Scale assessments indicates substantial attention to fall-risk screening. The facility should connect screening results to action through standardized prevention bundles, environmental risk checks, assisted-mobility protocols, and post-fall reviews for high-risk patients.

### 3. Make social-needs assessment actionable

Assessment of Health and Social Care Needs is the highest-volume procedure shown, at approximately 4,596 records. The hospital should establish a closed-loop referral process so identified needs are linked to case management and appropriate community resources.

### 4. Expand behavioral-health screening and referral capacity

Depression screening appears among the highest-volume procedures. The hospital should ensure positive screens trigger timely clinical assessment, referral, follow-up, and documentation rather than functioning only as a screening activity.

### 5. Standardize medication reconciliation

Medication reconciliation is one of the prominent procedures in the dashboard. The facility should prioritize reconciliation at admission, transitions of care, and discharge, with pharmacist or trained clinical review for higher-risk patients and clear escalation for discrepancies.

### 6. Use historical admission peaks for capacity planning

Admissions peak in 2014 in the displayed historical trend. Hospital leadership should use this peak as a stress-test benchmark for staffing, beds, emergency/acute-care capacity, and support services.

### 7. Investigate high utilization and repeat encounters

The dashboard shows 27,891 encounters for 974 displayed patients, equivalent to approximately 28.6 encounters per patient at the aggregate level. The facility should identify high-utilization cohorts and assess whether frequent encounters are driven by chronic disease, inadequate post-discharge support, behavioral-health needs, social barriers, or potentially avoidable utilization.

### 8. Develop a targeted uninsured-patient strategy

The payer distribution shows a large No Insurance segment. The hospital should strengthen financial counseling, eligibility screening, benefits navigation, charity-care assessment, and referral to available coverage programs to improve access and reduce avoidable financial exposure.

### 9. Align workforce and service capacity with the patient mix

Because the observed population is heavily senior-oriented and includes high volumes of screening, medication, social-needs, and safety assessments, workforce planning should

consider geriatric expertise, nursing capacity, case management, pharmacy support, behavioral health, and social work resources.

## 10. Establish an outcomes-based management program

The facility should move from tracking procedure volume alone to measuring whether interventions improve outcomes. Leadership can monitor falls, readmissions, length of stay, emergency revisits, medication discrepancies, referral completion, uninsured conversion, and patient experience.

### Recommended leadership priority

The strongest near-term opportunity is to combine senior-focused care, fall prevention, medication reconciliation, social-needs intervention, and targeted management of high-utilization patients. These priorities are directly connected to the dominant population and highest-volume activities visible in the dashboard.

## 5. Impact / Business Value

The value of this analysis is not simply better reporting; it is the ability to direct hospital resources toward areas most likely to affect patient outcomes, operational efficiency, and financial sustainability.

| Value Area               | Dashboard finding                                            | Facility opportunity                                               | Suggested outcome KPI                 |
|--------------------------|--------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------|
| Patient safety           | Senior-heavy population + high fall-risk assessment activity | Reduce inpatient falls and fall-related harm                       | Falls per 1,000 patient days          |
| Care quality             | High-volume medication reconciliation + depression screening | Reduce medication discrepancies and improve follow-up              | Discrepancy rate; referral completion |
| Care coordination        | Large volume of social-needs assessments                     | Close the loop between identified needs and services               | Social-referral completion            |
| Capacity                 | Historical admissions peak in 2014                           | Improve staffing and bed/resource readiness                        | Occupancy; staffing-to-demand ratio   |
| Utilization              | 27,891 encounters across 974 displayed patients              | Identify avoidable repeat utilization and high-risk cohorts        | 30/90-day revisit rate                |
| Financial sustainability | Large No Insurance segment                                   | Improve coverage navigation and reduce uncompensated-care exposure | Uninsured rate; coverage conversion   |

Portfolio-ready impact statement: Used historical patient-record analytics to identify a senior-dominant population, peak admission demand, high-volume clinical assessments, repeat-utilization signals, and a substantial uninsured segment—then translated those findings into hospital-level recommendations covering patient safety, care coordination, capacity, utilization management, and financial sustainability.

### How the hospital could measure impact

- Establish a baseline for each recommended intervention using the current reporting period.
- Set 90-day and 12-month targets for safety, quality, utilization, and financial indicators.

- Assign an accountable clinical or operational owner to each intervention.
- Review results by age group, payer, procedure, and utilization cohort so subgroup effects are visible.
- Use pre/post and, where feasible, comparison-group analysis before attributing changes to an intervention.

## 6. Tech Stack & Excel Technical

### Tools and features used

| Technology                       | Role in solution                                                                                                                                             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Microsoft Excel (desktop)</b> | Data preparation, analysis, dashboard development, and visual design. Saved as .xlsb to keep the large workbook manageable.                                  |
| <b>Source data sheets</b>        | Five tables (encounters, patients, procedures, payers, organizations) plus a data_dictionary sheet that defines every field and the keys linking the tables. |
| <b>PivotTables</b>               | Summarize the source data behind the KPI cards and charts.                                                                                                   |
| <b>Charts</b>                    | Line, bar, and column charts built from the PivotTables to show admissions, top procedures, age and gender mix, and payer volumes.                           |
| <b>Slicers and Timeline</b>      | Age Group, Gender, and Payer slicers plus a year Timeline filter every KPI and chart at once.                                                                |
| <b>Excel formulas</b>            | Lookup and logical formulas such as XLOOKUP and IFS (with UNIQUE and CONCAT) to link tables and prepare fields.                                              |

### Workbook structure

| Sheet                                                          | Purpose                                                                                        |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Dashboard</b>                                               | Final interactive dashboard with KPI cards, charts, slicers, timeline, and Key Insights panel. |
| <b>Pivots</b>                                                  | PivotTables that feed the dashboard visuals.                                                   |
| <b>data_dictionary</b>                                         | Field definitions for every source table.                                                      |
| <b>patients, procedures, payers, organizations, encounters</b> | Source data tables.                                                                            |

### Data preparation workflow

1. Review the source tables and the data dictionary to understand each field and the keys that link the tables.
2. Bring related fields together, such as payer and patient attributes, using lookup formulas like XLOOKUP.

3. Create derived fields such as Age Group (Adult 18-64 and Senior 65+) using logical formulas like IFS.
4. Build PivotTables on the Pivots sheet for each KPI and chart.
5. Create charts from the PivotTables and arrange them on the Dashboard sheet.
6. Connect the slicers and timeline to the PivotTables so one selection filters every visual.

### Dashboard design and interaction

- KPI cards establish the executive headline before the user reads the charts.
- Slicers and a timeline for year, age group, gender, and payer provide focused segmentation.
- The admissions line chart answers the 'when' question, while procedure ranking answers the 'what' question.
- Age/gender and payer visuals provide population and coverage context.
- The Key Insights panel reduces cognitive load by converting visual observations into plain-language takeaways.
- Possible next iteration: add separate views for patient, encounter, procedure, payer, and time period.

### Best practices and next steps

- Keep raw data, PivotTables, and the dashboard on separate sheets so each layer is easy to maintain.
- Use Data > Refresh All after the source data changes so every PivotTable and chart updates.
- Connect slicers to all relevant PivotTables so one selection filters the whole dashboard consistently.
- Validate dashboard totals against the source tables before sharing results.
- Keep field definitions in a data dictionary so others can understand and reuse the workbook.

## Reference Notes

The dashboard values and visual interpretations in this case study come from the Excel dashboard (Patient\_Dashboard.xlsx).

### Portfolio presentation tip

When presenting this project in an interview, walk through it as a business story: the problem → the data → the KPI layer → the key findings → the action enabled by each finding. Then open the Excel workbook to demonstrate the slicers, timeline, PivotTables, and formulas.